

FINAL YEAR RESEARCH PROJECT

End Term Presentation



A Modular Vision-Language Framework for Verifiable Image Captioning and Visual Dialogue Using Structured Scene Facts

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Presentation Outline

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Problem Statement

- Despite rapid progress in image captioning and visual question answering, current systems still struggle with a fundamental issue: they often generate descriptions that are fluent but not necessarily accurate.
- Many modern models rely heavily on learned patterns from large datasets, which makes them good at sounding convincing, yet they frequently “hallucinate” objects or relationships that are not actually present in the image.
- This becomes a serious limitation in real-world applications where correctness matters more than linguistic elegance.
- For example, in safety-critical domains, even a small misinterpretation of spatial relationships or object presence can lead to misleading conclusions.
- The core challenge, therefore, lies in ensuring that generated captions are not just plausible, but explicitly grounded in verifiable visual evidence.

Motivations

- The motivation behind this work stems from the need to bridge the gap between perception and language in a more reliable way.
- Humans naturally describe scenes based on what they actually observe, including spatial arrangements and interactions between objects.
- Replicating this behavior in machines requires moving beyond purely generative approaches toward systems that first understand and structure the visual scene before describing it.
- By grounding language generation in explicit scene facts such as object positions, depth, and interactions; it becomes possible to create captions that are both interpretable and trustworthy.
- This approach not only improves accuracy but also introduces an important capability: the system can acknowledge uncertainty instead of fabricating information, making it far more suitable for practical and high-stakes applications.

Literature Review

Table 1: Summary of Existing Approaches and Research Limitations

Approach / Model	Author	Core Idea	Strengths	Limitations	Research Outcome
CNN-RNN Encoder-Decoder Models	Oriol Vinyals et al. (2015)	Use CNNs for feature extraction and RNNs for caption generation	Simple and effective baseline; captures basic object information	Produces generic captions; lacks contextual and spatial understanding	Established the foundational end-to-end neural pipeline for automated image captioning.
Attention-based Models	Kelvin Xu et al. (2015)	Introduce attention to focus on relevant image regions	Improved caption quality and relevance	Still limited in modeling complex relationships between objects	Proved that dynamic spatial feature weighting improves descriptive accuracy and evaluation metrics (BLEU/CIDEr).
Transformer-based Models	Ashish Vaswani et al. (2017)	Use self-attention for better sequence modeling	Generates fluent and diverse captions	Weak grounding; prone to hallucination	Revolutionized sequence modeling, enabling massively parallel training and state-of-the-art linguistic fluency.
CLIP-based Vision-Language Models	Alec Radford et al. (2021)	Align visual and textual embeddings using large-scale pretraining	Strong semantic understanding; zero-shot capabilities	Does not explicitly model spatial relationships	Created universal visual-semantic embeddings, pioneering zero-shot classification and text-to-image alignment.

Literature Review

Table 2: Summary of Existing Approaches and Research Limitations

Approach / Model	Author	Core Idea	Strengths	Limitations	Research Outcomes
LLaVA / Multimodal LLMs	Haotian Liu et al. (2023)	Combine vision encoders with large language models	Human-like responses; supports dialogue and VQA	Often hallucinates details; lacks factual grounding	Pioneered instruction-tuned visual dialogue, standardizing the modern conversational VLM paradigm.
Scene Graph-based Methods	Justin Johnson et al. (2015)	Represent objects and their relationships explicitly	Improves interpretability and relational reasoning	Requires annotations; limited scalability and depth reasoning	Demonstrated that explicit symbolic structures (nodes/edges) drastically improve relational and attribute accuracy.

Research Gap & Improvements

- Existing image captioning models improve fluency but often lack grounding, leading to hallucinated objects and incorrect relationships.
- Attention and scene graph methods provide structure but fail to capture full spatial context or require heavy annotations.
- Proposed approach introduces fact-grounding using object segmentation, depth estimation, and structured scene representation.
- Ensures more accurate, interpretable, and reliable captions by constraining generation with visual evidence and handling uncertainty.

Proposed Solution & Architecture

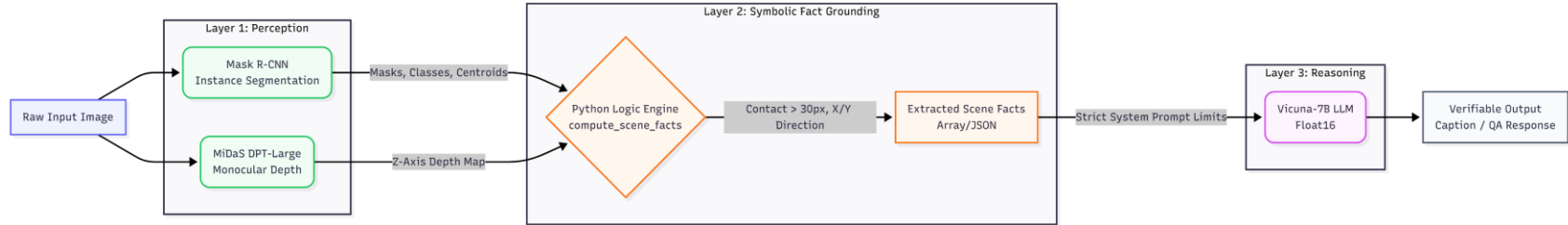


Fig. 1: Workflow of the Proposed Model

Proposed Solution & Architecture

cont.

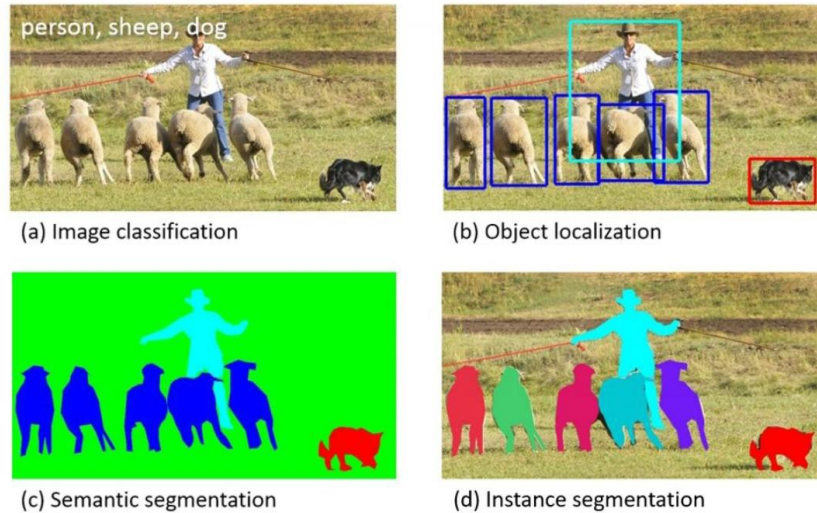


Fig 2: R-CNN segmentation

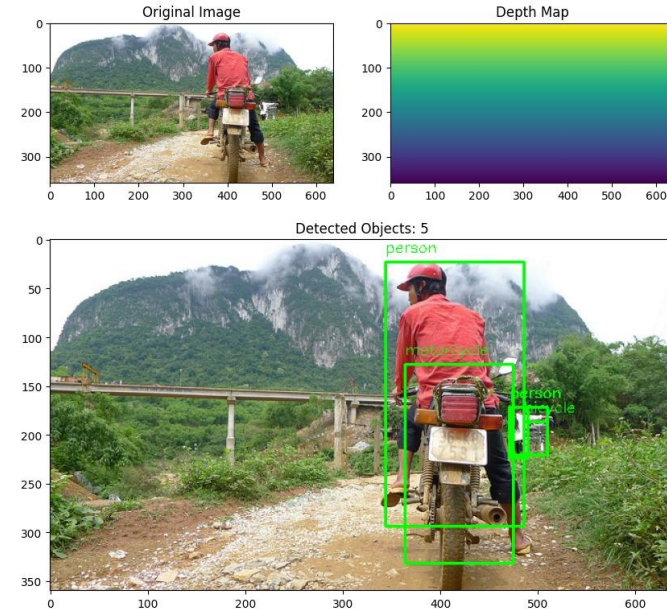


Fig 3: Depth Analysis through MiDaS

Dataset & Tools/Technologies

Category	Name	Purpose / Role	Key Features
Dataset	MS COCO 2017	Training and evaluation of image captioning models	Large-scale dataset with images and multiple captions; diverse object categories
Dataset	Flickr8k	Benchmark dataset for caption generation	Smaller dataset; useful for quick prototyping and validation
Object Detection	Mask R-CNN	Instance segmentation and object extraction	Provides precise object masks and class labels
Depth Estimation	MiDaS	Estimation of relative depth from images	Enables understanding of spatial relationships (near/far)
Vision Encoder	CLIP	Extract visual embeddings aligned with text	Strong zero-shot capabilities; semantic understanding
Language Model	Vicuna-7B	Caption generation and VQA	Conversational LLM; generates coherent, human-like text
Framework	PyTorch	Model development and training	Flexible deep learning framework with GPU support
Environment	Google Colab / Kaggle	Experimentation and training platform	Access to GPUs/TPUs; easy dataset integration

Progress in Implementation Plan & Methodology

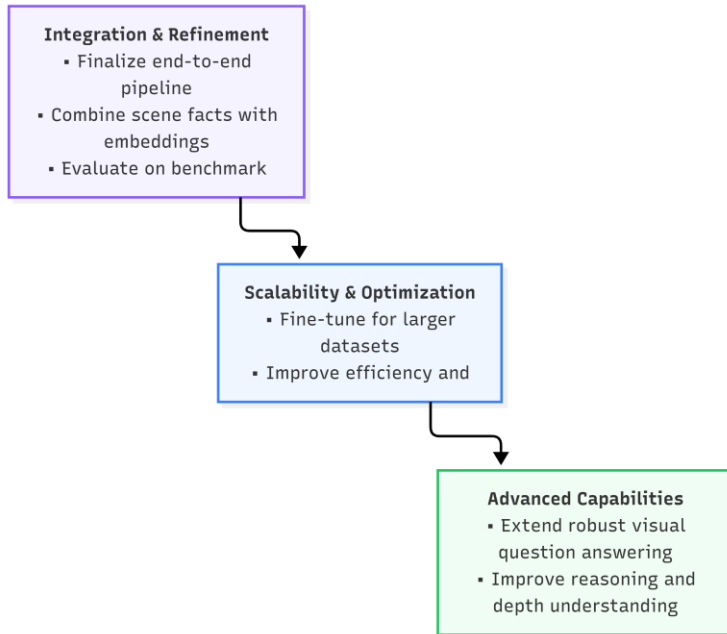
- Current work focuses on building the encoding pipeline, forming the system's foundation.
- Involves dataset preprocessing and integrating object detection, segmentation, and depth estimation.
- Components are being tested for consistency and effective interaction.
- Scene graph design is under exploration to refine relationships and move toward a stable fact-grounded captioning system.



Fig 4: detectron2 object detection (image credit- detectron)

Future Work Plan

- Next phase focuses on integrating all components into a unified end-to-end pipeline.
- Combine scene facts with vision-language embeddings to improve grounding and coherence.
- Evaluate performance on datasets like MS COCO for accuracy and reduced hallucination.
- Future work includes fine-tuning, extending to VQA, and improving efficiency and scalability.



References

1. Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common Objects in Context. In European Conference on Computer Vision (ECCV) (pp. 740–755). Springer.
2. He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask R-CNN. In IEEE International Conference on Computer Vision (ICCV) (pp. 2961–2969).
3. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., et al. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In International Conference on Learning Representations (ICLR).
4. Ranftl, R., Bochkovskiy, A., & Koltun, V. (2021). Vision Transformers for Dense Prediction (MiDaS).
5. Lin, T. Y., Maire, M., Belongie, S., et al. (2014). Microsoft COCO: Common Objects in Context. European Conference on Computer Vision (ECCV).
6. Xu, K., Ba, J., Kiros, R., et al. (2015). Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. ICML

*Thank
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